Version 3.0

Anti-Money Laundering (AML) Monitoring Compliance Guidance (March 2026)

This document presents a comprehensive framework for Anti-Money Laundering (AML) transaction monitoring using Jube, an open-source platform designed for real-time fraud detection and behavioural analytics. The framework aligns with guidance from the Financial Action Task Force (FATF) and the Wolfsberg Principles and integrates a data-driven, risk-based methodology supported by quantitative modelling, feature engineering, and defensible threshold calibration.

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Amendments

Date	Author	Version	Description
28th February 2025	Richard Churchman	2.0	Substantial update and rewrite from version one. Improved writing and linked to documentation instead of using images.
3rd March 2025	Richard Churchman	2.1	Updated to include details about training.
6th February 2026	Richard Churchman	3.0	Updated for more rigorous analytical methodology.

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Introduction to Methodology

This document outlines a framework designed to assist in monitoring compliance with Anti-Money Laundering (AML) regulations using Jube, open-source software utilized for fraud prevention and transaction monitoring, but with a sharp focus on regulatory prescribed transaction monitoring such as Anti Money Laundering (AML). This framework is based on regulatory guidance from the Financial Action Task Force (FATF), and the Wolfsberg Principles. Jube, being open-source and with global reach, is foreseeably exposed to local market regulations, and the working assumption is that such regulations will be derivative of FATA and the Wolfsberg Principles to some extent, but not enough to be compliant.

The scope of AML compliance is extensive, and this document focuses specifically on the monitoring requirements outlined in the guidance. While guidance does not endorse a rigid, unthinking approach, it highlights several clear risk factors, including:

- Country and Geographic Risk
- Distribution Channels Risk
- Transaction Behavioural Risk
- Product Liquidity and Use of Cash-Like Instruments (e.g., utility instruments like petrol, highly negotiable instruments like Bitcoin, and products linked to crime such as gambling and adult services)
- Anonymous Financial Products
- Online Products Vulnerable to Impersonation
- Person-to-Person Remittance and Low Financial Inclusion Products

The guidance acknowledges that risk factors often interact, with high-risk factors potentially being offset by low-risk ones. While the guidance suggests a subjective, experience-based weighting of risks, this approach is prone to judgmental bias. A data-driven approach, combined with subjective judgment, is recommended as a more reliable method. This trade-off approach aligns well with supervised and unsupervised learning techniques.

Training

Speed up implementation with hands-on, face-to-face training from the developer.

While transaction monitoring encompasses a broad spectrum of activities, this training specifically focuses on the AML (Anti-Money Laundering) use case. The training is designed to support compliance efforts and is structured around this guidance. This guidance is intended to be adapted by compliance managers to align with your organization's specific regulatory obligations, which are often derived from Financial Action Taskforce (FATF) guidelines and further reflected in the Wolfsberg Principles.

It is important to note that Jube operates as a real-time system. As a result, while the training emphasizes implementation of this guidance, other transaction monitoring use cases will also be comprehensively addressed by implication. The underlying concepts are largely adjacent and require similar methodologies, albeit with subtle differences in application.

Visit www.jube.io/jube-training for more information.

Core Principles of AML Monitoring

The overarching principle of AML compliance is Know Your Customer (KYC), achieved through due diligence, understanding the business and products, and conducting ongoing transaction monitoring in a timely (though not necessarily real-time) manner. Financial institutions are granted significant latitude under a Risk-Based Approach (RBA), allowing them to assess risk factors and implement controls proportionate to the risks posed by their products.

Countermeasures and Customer Due Diligence (CDD)

The primary countermeasure available to financial institutions is the rigorous application of Customer Due Diligence (CDD). CDD aims to ensure that institutions can verify the identity of customers, beneficial owners, and the nature of their transactions. There are five practical levels of CDD:

- Anonymous: No due diligence.
- Simplified Due Diligence (SDD): Verification of identity, sanctions, political exposure, and address. Monitoring product use within strict thresholds and limits.

- Enhanced Due Diligence (EDD): Verification of identity, address, and source of funds.
- Enhanced Monitoring (EM): Diligent monitoring of transactions within tight thresholds or manually.
- Suspicious Activity Reporting (SAR): Reporting suspicious activities to financial crime investigation authorities.

The weaker the CDD, the greater the exposure to money laundering. Therefore, the standard of CDD must be considered during monitoring, alongside a thorough understanding of product risks and prevailing risk factors. Additionally, maintaining a robust audit trail is critical to demonstrate diligent transaction monitoring and CDD escalation to regulators.

Automated Monitoring Systems

The use of automated monitoring systems is permitted and, in some cases, encouraged by regulatory guidance. However, the Money Laundering Reporting Officer (MLRO) must fully understand how these systems function. Given the evolving nature of financial crime, monitoring frameworks must be adaptable to emerging threats.

Regarding "tipping off" (the risk of alerting suspects to investigations), there is no requirement for real-time monitoring. Implementing systems on a batch or asynchronous basis (after the event) can reduce the risk of compromising investigations. Notwithstanding, Jube is real-time in its nature, refereeing to data integration as follows in this document.

Jube’s Data-Driven Risk-Based Approach

Jube promotes a Data-Driven Risk-Based Approach to monitoring, addressing a key challenge in AML: the absence of absolute outcomes. Instead, AML outcomes involve varying levels of suspicion. To address this, the proposed model focuses on anomaly detection and classification based on subjective levels of suspicion, as determined by rules that escalate cases to case management.

This approach ensures that monitoring is both adaptive and proportionate, leveraging data-driven insights to enhance the effectiveness of AML compliance efforts.

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About Jube

Jube is an open-source software designed for monitoring transactions and events. It features:

- Real-Time Data Processing
- AI-Driven Decision-Making
- Case Management

Jube excels in fraud prevention and abuse detection, with a particular focus on the Anti-Money Laundering (AML) transaction monitoring use case. The platform is built on the principle that adjacent use cases—while keeping in mind Jube's fastidious real-time capabilities—will derive from the same foundational methodology.

Focus on AML Transaction Monitoring

Jube maintains a sharp focus on the AML transaction monitoring use case, developing deep expertise in this area. The platform's training and support are strongly oriented toward this use case and its associated methodology.

Value Proposition for End Users

It is recognized that the discipline of AML is well-established. When end users evaluate Jube, they are typically not looking for a greenfield implementation but rather:

- Cost Reduction: Eliminate the prohibitive costs associated with proprietary solutions.
- Vendor Lock-In Eradication: Gain flexibility and control by avoiding dependency on single vendors.
- Improved Compliance: Access more advanced tooling to enhance compliance with regulatory requirements.

Jube represents a powerful, cost-effective, and flexible solution for AML transaction monitoring. By leveraging its real-time capabilities, advanced tooling, and open-source nature, financial institutions can reduce costs, eliminate vendor lock-in, and enhance compliance. Jube's focus on AML, combined with its adaptability to derivative use cases, makes it an asset in the fight against financial crime.

Further Reading

- About Jube: https://jube-home.github.io/jube/
- Getting Started: https://jube-home.github.io/jube/GettingStarted/
- GitHub: https://github.com/jube-home/jube

Risk Factors

Risk factors are developed by aggregating customer account transactions and product characteristics. These factors focus on four core areas, aligned with the most prevalent risk factors outlined in regulatory guidance:

Characteristic Grouping	Description
Product	This grouping captures the behavioural characteristics of financial products. It includes summary statistics that describe typical product behaviour and the degree of variation in that behaviour.
Product MCC Spend	This grouping contains aggregated data on spending by Merchant Category Code (MCC) for a given product type. It provides summary statistics describing typical monetary outflows for the product.
Product Country Spend	This grouping contains aggregated data on spending by Country Code for a given product type. It provides summary statistics describing typical monetary outflows for the product.
Account Characteristics	This grouping captures behavioural characteristics of individual accounts. It includes summary statistics and behavioural flags that describe account behaviour and the operating environment of the account.

Modelling Risk Factors in Jube

The following factors will be modelled in Jube to enable Machine Learning and Activation Rules. In Jube, risk factors can be modelled using Abstraction Rules. Below is an explanation of the factors within their core groupings:

Sanctions

- Distance: First and foremost, the obligation to screen customer names, source and destination account names via sanctions lists. Distance algorithms ensure than minor variations, deliberate or otherwise, do not evade sanctions screening.

Product Characteristics

- Typical Behaviour: Summary statistics describing the normal usage patterns of the product (e.g., average transaction size, frequency, and volume).
- Behavioural Variance: Measures of how much the product's usage deviates from the norm (e.g., standard deviation of transaction amounts or frequencies).

Product MCC Spend

- MCC-Based Spending Patterns: Aggregated data on spending by Merchant Category Code (MCC), such as retail, travel, or entertainment.
- Typical Outflows: Summary statistics describing the average monetary outflows for specific MCCs associated with the product.

Product Country Spend

- Country-Based Spending Patterns: Aggregated data on spending by Country Code, highlighting geographic trends in product usage.
- Typical Outflows: Summary statistics describing the average monetary outflows for specific countries associated with the product.

Account Characteristics

- Behavioural Flags: Indicators of unusual account activity (e.g., sudden spikes in transaction volume, changes in spending patterns).
- Environmental Factors: Contextual information about the account's operating environment (e.g., high-risk jurisdictions, use of anonymous products).
- Summary Statistics: Metrics describing the account's transaction behaviour (e.g., average balance, transaction frequency, and amounts).

By modelling these risk factors, Jube enables a data-driven approach to AML monitoring, allowing financial institutions to identify anomalies, assess risks, and implement targeted controls effectively. This structured approach ensures compliance with regulatory requirements while minimizing judgmental bias and enhancing the overall effectiveness of transaction monitoring.

Lists

The Data-Driven Risk-Based Approach relies on several lists that can be referenced in Activation Rules and Machine Learning models. These lists require regular maintenance to remain effective. Below are the key lists that should be included and regularly updated by the MLRO:

List Name	**Description**
FATF Uncooperative Countries	A list of countries identified by the Financial Action Task Force (FATF/GAFI) as non-cooperative in AML efforts.
High-Risk Human Trafficking Countries	A list of countries identified as high-risk for human trafficking, based on resources such as the UNODC Global Report on Trafficking in Persons.
Low GDP Per Capita Countries	A list of countries with low GDP per capita, used to identify high spending from regions with limited economic means.
Political Instability Countries	A regularly updated list of countries exhibiting political instability risks.
Sanctioned Countries	A list of countries subjected to international sanctions.

Further Reading

- Lists: https://jube-home.github.io/jube/Configuration/Models/Lists/
- Dictionaries: https://jube-home.github.io/jube/Configuration/Models/Dictionaries/

Sanctions

The primary requirement is to screen customer names, as well as the names of source and destination accounts, against sanctions lists. To account for minor variations—whether

intentional or unintentional—distance algorithms (e.g., Levenshtein Distance) are used to ensure effective screening.

Characteristic	Definition	Period Grouping
Sanctions Distance Recipient on Send	Sanctions match with a Levenshtein Distance greater than two, applied to the recipient's name for each transaction.	Period: For each transaction
Sanctions Distance Sender on Receipt	Sanctions match with a Levenshtein Distance greater than two, applied to the sender's name for each transaction.	Period: For each transaction

Sanctions could be taken to be any textual dataset that has been published via a regulator or central bank, such as US Office of Foreign Assets Control, or more universally available as CSV from consolidated sources such as OpenSanctions.

Further Reading

- Sanctions: https://jube-home.github.io/jube/Configuration/Sanctions/

Product Characteristics

These characteristics are available for each **Product ID** and are used to describe the typical behaviour and variability of financial products. Below is a detailed breakdown of each characteristic, including its definition, applicable periods, and grouping:

Characteristic	Definition	Period Grouping
Average Inflow	The average amount deposited into a product.	Periods: Day, Week, Month, Year, Lifetime. Grouped: POS and Cash/Quasi Cash.
Variance Inflow	Describes the variability of the average inflow, indicating its reliability.	Periods: Day, Week, Month, Year, Lifetime. Grouped: POS and Cash/Quasi Cash.
Average Outflow	The average amount withdrawn from a product.	Periods: Day, Week, Month, Year, Lifetime. Grouped: POS and Cash/Quasi Cash.
Variance Outflow	Describes the variability of the average outflow, indicating its reliability.	Periods: Day, Week, Month, Year, Lifetime. Grouped: POS and Cash/Quasi Cash.
Average Maximum Inflow	The average of the maximum deposit amounts across accounts for the product.	Periods: Day, Week, Month, Year, Lifetime. Grouped: POS and Cash/Quasi Cash.
Variance Maximum Inflow	Describes the variability of the average maximum inflow, indicating its reliability.	Periods: Day, Week, Month, Year, Lifetime. Grouped: POS and Cash/Quasi Cash.

Characteristic	Definition	Period Grouping
Average Maximum Outflow	The average of the maximum withdrawal amounts across accounts for the product.	Periods: Day, Week, Month, Year, Lifetime. Grouped: POS and Cash/Quasi Cash.
Variance Maximum Outflow	Describes the variability of the average maximum outflow, indicating its reliability.	Periods: Day, Week, Month, Year, Lifetime. Grouped: POS and Cash/Quasi Cash.
Average Minimum Inflow	The average of the minimum deposit amounts across accounts for the product.	Periods: Day, Week, Month, Year, Lifetime. Grouped: POS and Cash/Quasi Cash.
Variance Minimum Inflow	Describes the variability of the average minimum inflow, indicating its reliability.	Periods: Day, Week, Month, Year, Lifetime. Grouped: POS and Cash/Quasi Cash.
Average Minimum Outflow	The average of the minimum withdrawal amounts across accounts for the product.	Periods: Day, Week, Month, Year, Lifetime. Grouped: POS and Cash/Quasi Cash.
Variance Minimum Outflow	Describes the variability of the average minimum outflow, indicating its reliability.	Periods: Day, Week, Month, Year, Lifetime. Grouped: POS and Cash/Quasi Cash.
Average Frequency Inflow	The average frequency (count) of deposits across accounts for the product.	Periods: Day, Week, Month, Year, Lifetime. Grouped: POS and Cash/Quasi Cash.
Variance Frequency Inflow	Describes the variability of the average frequency of inflows, indicating its reliability.	Periods: Day, Week, Month, Year, Lifetime. Grouped: POS and Cash/Quasi Cash.
Average Frequency Outflow	The average frequency (count) of withdrawals across accounts for the product.	Periods: Day, Week, Month, Year, Lifetime. Grouped: POS and Cash/Quasi Cash.
Variance Frequency Outflow	Describes the variability of the average frequency of outflows, indicating its reliability.	Periods: Day, Week, Month, Year, Lifetime. Grouped: POS and Cash/Quasi Cash.

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Further Reading:

- Abstraction Rules: https://jube-home.github.io/jube/Configuration/Models/AbstractionRules/
- TTL Counters: https://jube-home.github.io/jube/Configuration/Models/TTLCounters/
- Environment Variables: https://jube-home.github.io/jube/Concepts/EnvironmentVariables/

Product MCC Spend

These characteristics are available for each Product ID and Merchant Category Code (MCC), providing insights into spending patterns associated with specific product types and merchant categories. Below is a detailed breakdown of each characteristic, including its definition, applicable periods, and grouping:

Characteristic	Definition	Period Grouping
Average Transaction Amount	The average transaction amount for a given product and MCC.	Periods: Rolling Three Months.
Variance Transaction Amount	Describes the variability of the average transaction amount, indicating its reliability.	Periods: Rolling Three Months.

Product Country Spend

These characteristics are available for each Product ID and Country Code of Spend, providing insights into spending patterns associated with specific product types and geographic regions. Below is a detailed breakdown of each characteristic, including its definition, applicable periods, and grouping:

Characteristic	Definition	Period Grouping
Average Transaction Amount	The average transaction amount for a given product and country.	Periods: Rolling Three Months.
Variance Transaction Amount	Describes the variability of the average transaction amount, indicating its reliability.	Periods: Rolling Three Months.

CDD and Account Factors

These characteristics are available for each Account or Relationship ID, providing detailed insights into customer behaviour and due diligence status. Below is a detailed breakdown of each characteristic, including its definition, applicable periods, and grouping:

Characteristic	Definition	Period Grouping
eKYC Flag	A Yes/No (1/0) flag indicating if the customer has been verified by eKYC verification methods.	Periods: Lifetime.
Manual KYC Flag	A Yes/No (1/0) flag indicating if the customer has been verified by manual KYC verification methods.	Periods: Lifetime.
Days Since Account Open	The number of days elapsed since the account was opened.	Periods: Lifetime.
Days Since Account Transacting	The number of days elapsed since the account first started transacting.	Periods: Lifetime.

Characteristic	Definition	Period Grouping
Identity Duplication	A Yes/No (1/0) flag indicating if heuristics (e.g., name, address, date of birth, email, or device recognition) suggest duplication with another account.	Periods: Lifetime.
Inflow	The absolute amount deposited into the customer account.	Periods: Day, Week, Month, Year, Lifetime. Grouped: POS and Cash/Quasi Cash.
Outflow	The absolute amount withdrawn from the customer account.	Periods: Day, Week, Month, Year, Lifetime. Grouped: POS and Cash/Quasi Cash.
Maximum Transaction Inflow	The largest deposit into the customer account.	Periods: Day, Week, Month, Year, Lifetime. Grouped: POS and Cash/Quasi Cash.
Minimum Transaction Inflow	The smallest deposit into the customer account.	Periods: Day, Week, Month, Year, Lifetime. Grouped: POS and Cash/Quasi Cash.
Maximum Transaction Outflow	The largest withdrawal from the customer account.	Periods: Day, Week, Month, Year, Lifetime. Grouped: POS and Cash/Quasi Cash.
Minimum Transaction Outflow	The smallest withdrawal from the customer account.	Periods: Day, Week, Month, Year, Lifetime. Grouped: POS and Cash/Quasi Cash.
Transaction Frequency Inflow	The number of deposits into the customer account.	Periods: Day, Week, Month, Year, Lifetime. Grouped: POS and Cash/Quasi Cash.
Transaction Frequency Outflow	The number of withdrawals from the customer account.	Periods: Day, Week, Month, Year, Lifetime. Grouped: POS and Cash/Quasi Cash.
Source of Funds Underwritten	A Yes/No value indicating if a Source of Funds Underwriting Process has been completed.	Periods: Lifetime.
Spends Outside Top 10 MCC	Indicates if the customer spends outside the Top 10 Merchant Category Codes (MCCs) for the product.	Periods: Day, Week, Month, Year, Lifetime.

Characteristic	Definition	Period Grouping
Spends Outside Top 5 Countries	Indicates if the customer spends outside the Top 5 Countries for the product.	Periods: Day, Week, Month, Year, Lifetime.
Spends Outlier for MCC	Indicates if the customer's spending falls outside the range of 90% of customers for a given MCC.	Periods: Day, Week, Month, Year, Lifetime.
Spends Outlier for Country	Indicates if the customer's spending falls outside the range of 90% of customers for a given country.	Periods: Day, Week, Month, Year, Lifetime.

Data Integration

The effectiveness of risk factors in Jube relies on the seamless integration of relevant data sources. Integration is achieved through the concept of models, which are configured to process, aggregate, and analyse data for monitoring purposes. Below is an outline of the proposed models, integration methods, and additional functionalities:

Example Models

The following models are recommended to ensure comprehensive data integration:

- Payment scheme authorisations:
  - Purpose: Capture all authorization records for payment scheme or brand transactions.
  - Outcome: Provides a complete and enhanced view of transactions processed through these payment networks.
- Financial Transactions and Clearing:
  - Purpose: Capture all financial transactions representing inflows and outflows to an account.
  - Includes: Deposits, withdrawals, and abridged cleared payment scheme transactions.
- Customer Due Diligence (CDD) Events
  - Purpose: Capture events related to customer verification processes.
  - Examples: Address validation, identity validation, source of funds verification, sanctions matching, etc.

Model Configuration

Each model is configured via Jube's user interface, where key values and fields are specified. The Request XPath page is used to define:

- Additional Fields: Fields made available for rule creation.
- Search Keys: Designated fields that enable the model to perform aggregations as outlined in the risk factors section.

Integration Method

The proposed integration method involves the following steps:

- Real-Time or Batch Preparation: Prepare messages in real-time or relay data from a database or other latent data store.
- Message Compilation: Compile the data into a JSON message.
- Data Transmission: Relay the message to Jube via HTTP.
- Serial Processing: Even if the data is latent, it will be processed serially in ascending datetime order, simulating real-time transaction processing.

Optional IP Address Decoding

Where available, IP addresses are decoded to provide geographic and contextual information about the session. This is achieved using an Inline Script to call IP2Location, which returns details such as:

- Country
- Coordinates
- ISP (Internet Service Provider)

- Region
- Usage Type/Connection
- Net Speed
- City
- Domain
- IDD & Area Code
- ZIP Code
- Elevation
- Weather Station

Outcomes of Integration

The integrated data and risk factors can be utilized in the following ways:

- Comprehensive Data Coverage: Ensures all relevant transaction and customer data is captured and analysed.
- Enhanced Monitoring: Enables real-time or near-real-time processing of latent data.
- Geographic Insights: Decodes IP addresses to provide contextual information for risk assessment.
- Flexibility: Supports integration with external services for PEP and sanctions validation.
- Actionable Insights: Facilitates the creation of dashboards, machine learning models, and activation rules for proactive AML compliance.

Further Reading

- Models: https://jube-home.github.io/jube/Configuration/Models/Models/
- Request XPath: https://jube-home.github.io/jube/Configuration/Models/RequestXPath/
- Inline Scripts: https://jube-home.github.io/jube/Configuration/Models/InlineScripts/
- Gateway Rules: https://jube-home.github.io/jube/Configuration/Models/GatewayRules/

MLRO Dashboards

The framework outlined in this document emphasizes that AML (Anti-Money Laundering) monitoring must be an evolving process. To support this, it is critical that MLRO Dashboards provide timely, comprehensible, and visually intuitive reporting. These dashboards enable the Money Laundering Reporting Officer (MLRO) to gain a deep understanding of:

- Product Behaviour
- Customer Activity
- Territorial Risks
- Distribution Channels

The scope of MLRO reports should not be limited to the current framework but should also serve as a tool for proactively creating new controls and adapting to emerging risks.

Objectives of MLRO Dashboards

The primary objectives of MLRO dashboards are:

- Drive Innovation: Facilitate the creation of innovative control hypotheses to address emerging risks.
- Monitor Rule Performance: Profile the performance of rules and assess their proportional contribution to the Data-Driven Risk-Based Approach.
- Evolve Risk Factors: Ensure that risk factors and rules are continuously updated based on insights derived from the dashboard.

Design and Customization

MLRO Dashboards can be instantly designed and customized using Jube's Reporting Designer. This tool allows for the creation of tailored visualizations and reports that align with the MLRO's specific needs, ensuring flexibility and adaptability.

Benefits of MLRO Dashboards and Risk Factor Lists

The combination of MLRO Dashboards and Risk Factor Lists ensures that financial institutions can:

- Monitor and Adapt: Continuously evolve their AML monitoring frameworks based on real-time insights.
- Proactively Manage Risks: Identify emerging threats and implement targeted controls to mitigate them.
- Ensure Compliance: Maintain alignment with regulatory requirements and global best practices.

Further Reading

- Visualisations: https://jube-home.github.io/jube/Configuration/Visualisations/

Machine Learning

Jube supports the use of risk factors in both Machine Learning (ML) and Activation Rules. The platform offers two types of Machine Learning:

- Classification (Supervised Learning).
- Unsupervised Learning.

Classification (Supervised Learning)

- Purpose: Identifies accounts that conform to a known classification, such as "Suspected Money Laundering."
- Process: Analysts must diligently flag highly suspicious transactions in the Case Management System using the star rating functionality.
- Outcome: A curated sample of highly suspicious transactions is used to train Jube's embedded classification engine.
- Goal: Distils numerous risk factors into a score that indicates the likelihood of a transaction being related to money laundering.
- Use Case: The score is recalled for each transaction processed through Jube and can be used in Activation Rules to escalate accounts to Case Management if the score exceeds a predefined threshold.

Limitations of Classification:

- Rapidly Changing Risks: Classification models are less effective in environments where risk typologies evolve quickly, as they are better at detecting known patterns than emerging risks.
- Anomaly Detection: Classification models are less capable of identifying new or anomalous behaviour.

Unsupervised Learning

To counter the limitations of Classification, Jube employs One-Class Support Vector Machine (SVM), a technique that groups customers based on similar behaviours using the risk factors described in this document. Unlike Classification, Unsupervised Learning does not require transactions to be marked as suspicious to learn.

Benefits of Unsupervised Learning:

- Anomaly Detection:
    - Most customers fall into a small number of clusters representing typical behaviour.
    - Outlier clusters represent anomalous behaviour, which can be flagged for further investigation.
- Unbiased:
    - Anomalies are identified based on behavioural similarities without predefined labels.

Exhaustive Adaptation Concepts

Traditional risk management or anomaly detection is typically rule-based. Examples of rules might include:

- More than four transactions in one day.
- More than two declined transactions in one day.

- More than two different transactions in one day.
- More than five transactions at the same merchant in one day.
- Transactions more than twice (x2) the customer's average transaction.

While rules are effective, their derivation is often anecdotal, and thresholds are rarely adapted over time. This leads to:

- Rule Proliferation: The number of rules tends to grow, making them difficult to manage.
- Diminishing Efficacy: Rules often fail to adapt to new risk typologies, reducing their effectiveness.

Jube combines Supervised and Unsupervised Learning to create a robust and adaptive risk management system.

Key Steps in Jube's Machine Learning Process:

- Data Extraction: Data is extracted from Jube for Abstraction Rules, which include continuous values such as transaction counts, declined transactions, and customer averages.
- Anomaly Detection: Data is trained based on anomaly, where highly anomalous events are classified as fraudulent.
- Blending Feedback Data: Anomaly data is blended with specific feedback data tagged by analysts, which is lower in volume but highly salient.
- Neural Network Training: Multiple neural network topologies (e.g., hidden layers, activation functions) are trialled, evolved, and trained based on the data.

Outcome:

- A complex neural network model is created on a quantitative basis.
- The model identifies optimal topology using the Levenberg-Marquardt Backpropagation Algorithm.
- The system continuously adapts by retraining models in a champion-challenger framework, ensuring ongoing relevance and accuracy.

Exhaustive Training Algorithm

Jube's Exhaustive training algorithm follows these steps:

1. Variable Collation: Eligible variables (e.g., from Abstraction Calculations, TTL Counters, and Abstraction Rules) are collated.
2. Data Sampling: A sample of data is extracted from the archive, excluding class variables.
3. Statistical Analysis: Statistics are performed on the sample for Z-score normalization.
4. Normalization: Continuous variables are normalized, while binary variables remain unchanged.
5. Anomaly Detection: A One-Class Support Vector Machine is trained to detect anomalies.
6. Data Blending: Anomaly data is blended with feedback data tagged by analysts.
7. Model Training: Neural networks are trained exhaustively, with topology and variables selected randomly.
8. Performance Validation: Models are validated using testing data, and the most performant model is selected.
9. Monte Carlo Simulation: A Monte Carlo simulation is performed to analyse model sensitivity.

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Jube goes some way to automate a rigorous analytical process, which is detailed as follows. While Exhaustive in particular sets out to automate, the nature of the platform, and the regulatory compliance (i.e., how and why) means that some rigor in the analytical methodology is required, which is set out in Appendix A (Data Foundations & Minimum Dataset Requirements), Appendix B (Analytical Methodology) and Appendix C (Model Validation and Performance Assurance) of this document.

Further reading

- Exhaustive Adaptation: https://jube-home.github.io/jube/Configuration/ExhaustiveAdaptation/
- HTTP Adaptation: https://jube-home.github.io/jube/Configuration/Models/HTTPAdaptation/Index.html

Activation Rules

Activation Rules are used to escalate accounts and transactions into **Case Management** based on risk factors and machine learning outcomes. Below is a list of proposed Activation Rules:

Name	**Description**
High TXN Amount >= 4000	A transaction exceeds EUR 4000 in value.
Automated Fuel Dispenser (5542) >= 3txns in 1 day	More than three fuel dispensers used in a single day. (Applies to utility transactions.)
High Velocity >= 7 Txns in a day	More than seven transactions in a day.
High Risk MCC >= 4txns in 5 hours	More than four transactions in a high-risk MCC within five hours.
High Withdrawal >= 2000 in 2 days	More than 2000 EUR withdrawn from an account within two days.
High Risk MCC High Amount >= 1000 in 2 days	Transactions in high-risk MCCs (e.g., utilities or negotiable instruments) exceeding 1000 EUR in two days.
High Risk Country >= 2000 in 5 hours	More than 2000 EUR transacted in a high-risk country within five hours.
High Amount >= 3000 over 2 days	More than 3000 EUR transacted within two days.
3 Countries in 6 hours and 1 HR Country	More than three transactions in three different countries within six hours, with one high-risk country.
3 Countries in 1 day Not Ecommerce	More than three transactions in one day, excluding eCommerce transactions.
High Risk MCC >= 4 Txns in 6 hours	More than four transactions in high-risk MCCs within six hours.

Name	Description
Same Merchant >=3 txns in 1 day	Transactions where the same merchant is used more than three times in one day.
ATM >= 500.00 EUR/day on 3 consecutive days	ATM withdrawals exceeding 500 EUR per day for three consecutive days.
High Risk MCC >= 50,000 in 1 day	A high-risk MCC transaction exceeding EUR 50,000 in one day.
High Deposit >= €300 in 6 hours	More than 300 EUR deposited into an account within six hours.
Load then Refund in 1 day	A load followed by a refund observed within one day.
Refund on two consecutive days	Two refunds observed on two consecutive days.
Multiple Refunds >= 3 in 2 days	More than three refunds made within two days.

Further Reading

- Basic Activation Rules: https://jube-home.github.io/jube/Configuration/Models/BasicActivationRules/
- Response Elevation: https://jube-home.github.io/jube/Configuration/Models/ResponseElevation/
- TTL Counter Activation Rule Incrementation: https://jube-home.github.io/jube/Configuration/Models/TTLCounterActivationRuleIncrementation/
- Activation Watcher: https://jube-home.github.io/jube/Configuration/Models/ActivationWatcher/
- Activation Rule Notifications: https://jube-home.github.io/jube/Configuration/Models/ActivationRuleNotifications/
- Activation Rules Suppression: https://jube-home.github.io/jube/Configuration/Models/ActivationRulesSuppression/
- Rule Compilation Algorithm: https://jube-home.github.io/jube/Configuration/Models/RuleCompilationAlgorithm/
- Reprocessing: https://jube-home.github.io/jube/Configuration/Reprocessing/

Case Management

Case Management is the central hub where accounts flagged by rules are reviewed and investigated by analysts. In Jube, cases are organized by workstreams, and each case can have one of the following statuses:

- Refer to MLRO: Escalated to the Money Laundering Reporting Officer for further review.
- No Further Action: No suspicious activity detected; case closed.
- Pending EDD Documentation: Awaiting Enhanced Due Diligence (EDD) documentation.
- Pending CDD Documentation: Awaiting Customer Due Diligence (CDD) documentation.
- SAR Sent by MLRO: A Suspicious Activity Report (SAR) has been filed by the MLRO.

Case Audit

Jube provides robust Case Audit features to ensure transparency and compliance. These features allow regulators to review the lifecycle of a case on demand. Key audit functionalities include:

- Single Open Case per Account:
  - Only one case can be open for an account at any given time.
  - Once a case is closed, it can be reopened if necessary.
- Full Case Journal:
  - Only one case can be open for an account at any given time.
  - Once a case is closed, it can be reopened if necessary.
- Full Case Journal:
  - A complete record of all open and previously closed cases is maintained.
- Action Tracking:
  - Every action performed on a case—whether voluntary (e.g., adding notes) or involuntary (e.g., reviewing a case)—is recorded in the audit trail.
- Analyst Notes:
  - Analysts are required to add notes to explain the circumstances surrounding case creation and their actions during the investigation.

Case Uploads

As part of the investigation process, Enhanced Due Diligence (EDD) documentation may be required. Jube allows for the seamless upload and management of such documents:

- Document Upload: EDD documentation can be uploaded directly to the case.
- Centralized Access: All relevant documents are stored in a specific location for easy review.

Case Escalation

Cases may need to be escalated for further review by the MLRO or for the creation of a Suspicious Activity Report (SAR). Escalation in Jube is straightforward:

- Change Workflow Status:
  - Analysts can escalate a case by changing its workflow status to "Refer to MLRO".

- MLRO Review:
  - The MLRO can filter and view escalated cases for review.
- SAR Creation:
  - SARs can be created by completing a case form within Jube.
  - MLRO reports allow for the creation of template SARs, which can be emailed directly to the relevant authorities.

Further Reading

- Case Management: https://jube-home.github.io/jube/Configuration/CaseManagement/

Summary

This document presented a framework for monitoring Anti-Money Laundering (AML) compliance using Jube, an open-source software designed for fraud prevention and transaction monitoring. The framework is based on regulatory guidance from the Financial Action Task Force (FATF) and the Wolfsberg Principles, focusing on transaction monitoring and risk-based approaches to AML compliance. It integrates both the high-level AML operating model and the deeper quantitative methodology required for rule justification, model calibration, and regulatory defensibility.

Risk Factors

- Country and Geographic Risk: High-risk jurisdictions.
- Transaction Behavioural Risk: Unusual transaction patterns.
- Product Risk: Liquidity, cash-like instruments, and anonymous products.
- Customer Risk: Low financial inclusion, person-to-person remittances, and online impersonation risks.

Core Principles

- Know Your Customer (KYC): Central to AML compliance, involving due diligence and ongoing monitoring.
- Risk-Based Approach (RBA): Financial institutions assess risks and implement proportionate controls.
- Customer Due Diligence (CDD): Five levels of diligence, from anonymous to enhanced monitoring and suspicious activity reporting (SAR).

Automated Monitoring

- Jube supports real-time and batch processing for transaction monitoring.
- Monitoring is implemented through anomaly detection and classification using supervised and unsupervised machine-learning techniques.
- Activation rules escalate suspicious cases for further investigation.

Data-Driven Risk-Based Approach

- Jube uses risk factors such as product characteristics, transaction behaviour, and geographic spending patterns.
- Machine-learning models (e.g., One-Class SVM) identify anomalies and classify suspicious activities.

- Risk factors are modelled using abstraction rules and integrated into activation rules for case escalation.

Case Management

- Cases flagged by rules are reviewed by analysts, with statuses such as Refer to MLRO, No Further Action, or SAR Sent.
- Robust audit trails ensure transparency, and EDD documentation can be uploaded for further investigation.
- Escalation to the Money Laundering Reporting Officer (MLRO) and SAR creation are streamlined.

MLRO Dashboards

- Provide real-time insights into product behaviour, customer activity, and territorial risks.
- Enable initiative-taking risk management and adaptation to emerging threats.

Integration and Data Processing

- Jube integrates data from various sources (e.g., MasterCard/Visa authorizations, financial transactions, and CDD events).
- IP address decoding provides geographic and contextual insights for risk assessment.
- Data is processed serially, even in batch mode, to simulate real-time monitoring.

Analytical Methodology and Minimum Dataset Requirements (Appendix A)

- The methodology adopts a minimum-dataset principle, centred on the transaction journal, which provides the essential behavioural and financial descriptors needed for defensible feature engineering, anomaly detection, and threshold calibration.
- Additional artefacts (IP telemetry, device metadata, session attributes) enhance context but are not prerequisites; their absence must not delay regulatory compliance.
- The Analytical Methodology defines how features are engineered, how labels are used when reliable, and how fallback modelling (probabilistic, unsupervised, expert-guided calibration) ensures continuous detection capability where labels are incomplete or weak.

Performance Measurement and Validation (Appendix C)

- A comprehensive KPI framework supports governance, including FPR, TPR, Precision, Rule/Feature Coverage, Calibration, Operational Metrics, and Continuous Improvement Metrics.
- Statistical techniques include exploratory data analysis, Bayesian modelling, anomaly-detection frameworks, threshold calibration via ROC/cost curves, sensitivity analysis, and population-drift monitoring.
- Validation encompasses feature-level distributional analysis, discriminatory-power assessment, calibration checks, ranking stability, drift testing, and back-testing.
- Formal sign-off checkpoints ensure all analytical artefacts are reviewed by Risk, Compliance, Audit, and operations stakeholders, with versioned and time-stamped documentation retained for auditability.

Value Proposition

- Cost Reduction: Eliminates prohibitive costs of proprietary solutions.
- Vendor Lock-In Eradication: Open-source nature ensures flexibility.
- Improved Compliance: Advanced tooling enhances regulatory adherence.

Jube provides a powerful, flexible, and cost-effective solution for AML transaction monitoring. By combining real-time capabilities with a rigorous quantitative annex—covering data sufficiency, fallback methods, performance measurement, and validation—the framework enables institutions to enhance compliance, reduce costs, and adapt to evolving financial crime threats. The result is a fully integrated, data-driven, risk-based methodology that is transparent, defensible, and aligned with regulatory expectations.

Appendix A: Data Foundations & Minimum Dataset Requirements

A coherent and defensible AML monitoring programme must begin with clarity regarding the minimum viable data estate required to implement the methodology. In keeping with regulatory expectations and the principle of proportionality, the foundational requirement is the transaction journal, which—when properly engineered—provides a sufficiently rich basis for detecting anomalous behaviour and supporting both rule-based and probabilistic controls. The transaction journal encapsulates the essential behavioural and financial descriptors upon which risk factors are constructed, and, crucially, offers the temporal granularity required to develop, monitor, and validate controls in a manner that is reproducible, empirically grounded, and operationally practical.

While additional artefacts such as IP telemetry, device metadata, session attributes, and biometric signals may enhance contextual richness, they are explicitly not prerequisites for compliance or for the deployment of the methodology described herein. Such artefacts are best treated as desirable or opportunistic rather than foundational: their presence can refine feature engineering and improve discriminative power, but their absence must not be permitted to delay the implementation of mandated AML controls. A pragmatic approach is therefore required—one that emphasises the sufficiency of the transaction journal while allowing the broader data environment to mature in parallel, rather than conditioning regulatory progress on the existence of an idealised data lake or vendor-grade behavioural telemetry.

This pragmatic stance is particularly important in regulated environments where project timelines, supervisory scrutiny, and legacy architecture place constraints on data accessibility. The methodology is expressly designed to operate effectively within such constraints: comprehensive feature construction, anomaly detection, threshold calibration, and model validation can all be achieved using the minimum dataset, with incremental enhancement applied as and when new data sources become reliably available. In this way, the institution remains compliant, auditable, and operationally aligned, while retaining a clear and ordered path towards continuous improvement of the analytical estate.

Further Reading

- Jube R Advanced Analytics Guidance: https://jube.io/AdvancedAnalyticsWithRGuidance.pdf
- Jube R Advanced Analytics Training: http://advancedanalyticswithrtrainingplan.pdf/

Appendix B: Analytical Methodology

This document introduces the methodology for developing rules, models, and policies in anti-money laundering and transaction fraud monitoring across a variety of channels, with a sharp focus on regulated environments. Given regulated environments and the prospect of a Central Bank audit, it is of paramount importance that a methodology be followed to ensure all stakeholders coalesce and, above all, allow the data, thus policy, to speak for itself.

Regulatory guidance tends not to be highly prescriptive, and it is for the regulated entity to create rules and policies within the broad guidance; therefore, it is incumbent on the bank to adopt a quantitative approach such that, upon scrutiny, objective justification for all regulations and policies exists. While the guidance might not be wholly prescriptive, it is not devoid of a framing of risk factors, which will hereinafter be referred to as Features, and, alongside the consultant's own expertise in the domain, provides a solid platform for implementing the methodology set out in this document.

Label Data Availability and Fallback Methodology

In practice, the methodology prioritizes the use of reliable labelled data to inform supervised modelling and validate rules, ensuring that detection mechanisms are grounded in known outcomes. When such labels are incomplete, inconsistent, or unavailable, a robust fallback approach is applied, leveraging unsupervised analysis, probabilistic inference, and expert-guided calibration to identify anomalous or high-risk activity. As new and more reliable labels become available, the methodology continuously incorporates them to refine models, thresholds, and rules, maintaining a defensible, auditable framework that ensures regulatory compliance and operational effectiveness throughout the lifecycle of the monitoring program.

Ideally, the bank maintains a dataset of transactions labelled as fraudulent or non-fraudulent. Such labels significantly enhance supervised learning and rule validation. However, in practice, existing label data is often incomplete, inconsistent, or unavailable, especially in early-stage implementations.

It is therefore critical that the methodology accounts for both reliable and unreliable label scenarios, ensuring continuous and defensible risk monitoring.

Standard Use of Label Data

When trustworthy labels exist:

- Use them to train supervised Bayesian or probabilistic models.
- Validate feature importance, rule thresholds, and model calibration against known outcomes.
- Compute KPIs such as True Positive Rate (TPR), False Positive Rate (FPR), Precision, and ROC curves using labelled data.

Labels are also used to support rule recall testing, sensitivity analysis, and operational calibration.

Fallback Methodology for Weak or Absent Labels

When label data is incomplete or unreliable, the methodology employs robust fallback techniques:

- Unsupervised / Probabilistic Detection:

  - Bayesian networks or similar probabilistic models detect anomalous behavior without requiring labels.
  - Conditional probabilities are computed based on feature states, historical distributions, and transaction context.
- Feature-based Scoring:
  - Risk scores are derived from engineered features such as transaction amount, velocity, deviation from historical patterns, and behavioral signals.
  - Multiple features are combined probabilistically or via weighted aggregation to flag high-risk transactions.
- Expert-guided Calibration:
  - High-risk transactions identified by probabilistic or feature-based scoring are reviewed by domain experts.
  - Expert feedback informs threshold selection and refinement of rules for deployment.
- Iterative Validation:
  - As reliable label data becomes available over time, models, thresholds, and rules are recalibrated.
  - All adjustments are fully documented for auditability and regulatory review.

Governance and Audit Considerations

The fallback methodology ensures that no critical risk is ignored due to missing labels. All steps — feature derivation, probabilistic inference, expert oversight, and rule thresholding — are documented and reproducible, satisfying audit and regulatory requirements. KPI tracking (e.g., FPR, TPR, ROC curves, operational metrics) remains active even under fallback conditions, ensuring continuous monitoring and defensible reporting.

Key Performance Indicators (KPIs)

The following KPIs provide quantitative measures for evaluating rule and model performance, ensuring robust governance and defensible reporting:

KPI	**Definition**	**Purpose / Regulatory Relevance**
False Positive Rate (FPR)	Percentage of legitimate transactions flagged as suspicious	Monitors operational efficiency; controls unnecessary customer friction
Detection Rate / True Positive Rate (TPR)	Percentage of fraudulent transactions correctly flagged	Measures effectiveness of fraud/AML detection; supports compliance objectives
Precision / Positive Predictive Value (PPV)	Fraction of flagged transactions that are truly fraudulent	Balances efficiency and accuracy; informs resource allocation for investigation teams
Rule Coverage / Feature Coverage	Proportion of transaction types and portfolio segments monitored by rules/models	Ensures comprehensive risk coverage and regulatory defensibility
Model Calibration Metrics	Comparison of predicted vs. actual outcomes (e.g., expected vs. observed probability of fraud)	Ensures probabilistic predictions are accurate and interpretable; supports auditability

Case / Operational Metrics	Volume of flagged transactions, time-to-review, analyst workload	Tracks operational impact of rules and models; helps maintain sustainable workflows
Cost Curves / ROC Analysis	Receiver Operating Characteristic (ROC) curves, cost-benefit curves for thresholds	Supports optimization of thresholds and trade-offs between detection and false positives
Continuous Improvement Metrics	Improvements in detection or reduction in false positives over time	Demonstrates ongoing governance, learning, and adaptive methodology

KPIs should be tracked per channel and transaction type to account for portfolio heterogeneity, and periodically reviewed with compliance, audit, and risk teams.

Statistical and Analytical Methodology

All rules and models are grounded in quantitative, reproducible, and auditable statistical techniques:

- Feature Engineering:
  - Derive candidate features from transaction journal, IP/proxy, device, session, and other accessible sources.
  - Features include velocity, amount distributions, ratios, historical patterns, and behavioural signals.
  - Oftentimes this emulates features of another fraud management system, compensating for lack of underlying data availability for analysis.
  - Feature Engineering is laid out more prescriptively in the following document: https://jube.io/JubeAMLMonitoringComplianceGuidance.pdf
- Exploratory Data Analysis (EDA):
  - Univariate Analysis: Examine distributions, outliers, and anomalies per feature.
  - Bivariate / Multivariate Analysis: Identify relationships and dependencies between features.
  - Outliers are investigated and contextualized before model inclusion.
- Label Analysis and Feature Selection:
  - Labelled transaction data, when available, is used to inform feature importance.
  - In cases of incomplete labels, unsupervised / Bayesian inference techniques are used to probabilistically assess anomalous behaviour.
  - Feature selection balances predictive power, interpretability, and deployability within vendor systems.
- Modeling Framework:
  - Bayesian Networks form the primary probabilistic framework for both anomaly detection and prescriptive analytics.
  - Probabilities of anomalous or fraudulent behaviour are inferred conditional on feature states.
  - Where label data exists, supervised Bayesian models are used to refine detection probability estimates.
  - Models are validated against historical data to ensure robustness, calibration, and defensibility.
- Validation and Sensitivity Analysis:

    - Rule thresholds are calibrated using ROC curves, cost curves, confusion matrices, and expected vs. actual outcomes.
    - Sensitivity of rules/models to parameter changes is tested to ensure stability.
    - Performance metrics (KPIs) are recalculated across simulated scenarios to assess robustness under varied transaction volumes and risk profiles.
    - Any calibration adjustments are documented for audit purposes.
- Continuous Improvement:
    - Models, rules, and thresholds are reviewed periodically with new transaction data, feature improvements, and evolving threat patterns.
    - KPI trends over time inform adjustments, ensuring that probabilistic models remain aligned with real-world behaviour.

The methodology ensures that all rules and models are defensible, reproducible, and aligned with regulatory guidance, allowing auditors and compliance officers to verify the rationale, implementation, and performance quantitatively.

Further Reading

- Jube R Advanced Analytics Guidance: https://jube.io/AdvancedAnalyticsWithRGuidance.pdf
- Jube R Advanced Analytics Training: http://advancedanalyticswithrtrainingplan.pdf/

Appendix C: Model Validation and Performance Assurance

A challenge with AML and fraud regulations is that they are not very prescriptive yet carry significant financial penalty if not implemented properly. In keeping with this methodology, everything is approached from a quantitative basis (i.e., we don't look at that because it is in this dense section of the probability density, etc.). To model validation, while this remains nonprescription for AML and fraud regulations, techniques have evolved through several iterations of Basel, and that can be overlaid into this regulatory problem.

Model validation within this methodology is designed to provide a rigorous, proportionate, and auditable assurance framework aligned in principle with Credit Risk model governance. Validation encompasses feature-level distributional analysis, calibration of probabilistic outputs, assessment of discriminatory power, stability and drift monitoring, sensitivity testing, and outcome-based back-testing where labels exist. The approach explicitly recognizes that fraud and AML environments are adversarial and non-stationary, that labelled outcomes may be incomplete or delayed, and that expert judgement must complement statistical evidence. Accordingly, validation focuses on defensibility, transparency, and stability of model behavior over time, rather than spurious precision, ensuring that all models and rules remain fit for purpose, interpretable, and aligned with regulatory expectations.

Objectives of Model Validation

Model validation ensures that all probabilistic models, rules, and thresholds are statistically sound, stable over time, and fit for purpose within a regulated environment. The objectives are to:

- Demonstrate that models behave consistently with observed data.
- Ensure outputs are interpretable, calibrated, and defensible.
- Identify model degradation, instability, or unintended bias.
- Provide an auditable trail equivalent in spirit to Credit Risk model validation.

Validation is performed prior to deployment, at material change, and on a periodic basis thereafter.

Distributional Analysis and Feature Validation

Consistent with Credit Risk practice, validation begins at the feature level:

- Empirical distributions of all key features are examined (amounts, velocities, ratios, behavioral indicators).
- Candidate parametric distributions (e.g. log-normal, Poisson, exponential, and mixture distributions) are assessed where appropriate.
- Goodness-of-fit is evaluated using visual inspection (histograms, plots) and statistical measures were meaningful.
- Features exhibiting instability, extreme skew, or structural breaks are either transformed, segmented, or excluded.

This ensures that model inputs are well-behaved, interpretable, and stable, and that probabilistic inference is not driven by artefacts.

Model Calibration and Probability Validation

For probabilistic and Bayesian models:

- Predicted risk probabilities are compared to observed outcomes where labels exist (expected vs. observed).
- Calibration is assessed at portfolio, segment, and channel level.
- Where labels are weak or absent, calibration is assessed indirectly via:
    - Stability of posterior distributions
    - Consistency of anomaly scores over time
    - Expert review of high-risk tails

Calibration analysis ensures that model outputs can be interpreted as meaningful risk measures, not just rankings.

Discriminatory Power and Ranking Validation

Where labels exist, standard discriminatory measures are used:

- ROC curves and AUC.
- Precision / Recall.
- KS-style separation where applicable.

Where labels are weak or absent:

- Relative ranking stability is assessed across time windows.
- Concentration of risk in high-score bands is reviewed.
- Manual outcomes and downstream investigative feedback are incorporated.

This mirrors Credit Risk practice where full default histories may not yet exist.

Stability and Population Drift Analysis

Models and features are tested for temporal stability, including:

- Feature distribution drift over time.
- Changes in score distributions.
- Volume and composition of high-risk flags.

While PSI/CSI may be used cautiously, the emphasis is on interpretation rather than mechanical thresholds, reflecting the dynamic nature of fraud behavior.

Significant drift triggers:

- Review of feature definitions.
- Recalibration of thresholds.
- Potential model re-estimation.

Sensitivity and Stress Testing

Sensitivity analysis is conducted to understand model robustness:

- Impact of threshold shifts on FPR, TPR, and queue volumes
- Behavior under elevated transaction volumes or changed customer behavior
- Impact of removing or perturbing key features

This ensures that models are operationally safe and not brittle.

Back-Testing and Outcome Review

Where historical outcomes exist:

- Models and rules are back tested on prior periods.

- Differences between historical and current performance are analyzed.
- Any material deviations are explained and documented.

Back-testing supports:

- Regulatory defensibility.
- Management confidence.
- Change approval decisions.

Sign-Off Expectations

All validation activities produce documented artefacts. Recommended actions

At key stages throughout the methodology, formal sign-off is expected from designated stakeholders to ensure governance, accountability, and regulatory defensibility. These checkpoints provide assurance that all analyses, models, and rules are reviewed, validated, and approved before deployment. Typical sign-off points include:

- Feature Specification and Analytical Dataset Creation:
  - Approval by Risk and Compliance teams that the selected features are appropriate, relevant, and sufficient for the intended monitoring objectives.
- Modelling Framework and Threshold Definition:
  - Review and sign-off by Data Science Lead, Risk, and Compliance confirming the probabilistic model structure, assumptions, and threshold rationale.
- Rule Derivation and Deployment:
  - Approval by Operations, Risk, and Compliance ensuring that rules derived from model outputs are operationally feasible, aligned with risk appetite, and implementable within vendor systems.
- Validation, Sensitivity Analysis, and KPI Assessment:
  - Sign-off by Audit and Compliance to confirm that model performance, rule calibration, and KPI tracking meet regulatory and internal standards.
- Ongoing Review and Continuous Improvement:
  - Periodic governance reviews to approve updates to models, thresholds, or rules as new data, labels, or operational insights become available.

All sign-offs are documented, time-stamped, and retained as part of the audit trail. This ensures that all decisions are transparent, defensible, and fully traceable in compliance with regulatory expectations.

Further Reading

- Jube R Advanced Analytics Guidance: https://jube.io/AdvancedAnalyticsWithRGuidance.pdf
- Jube R Advanced Analytics Training: http://advancedanalyticswithrtrainingplan.pdf/